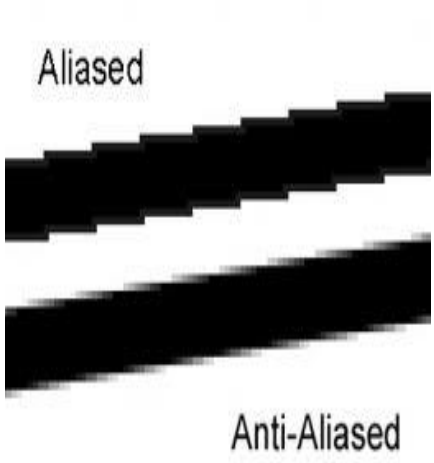
Lesson 12

Aliasing

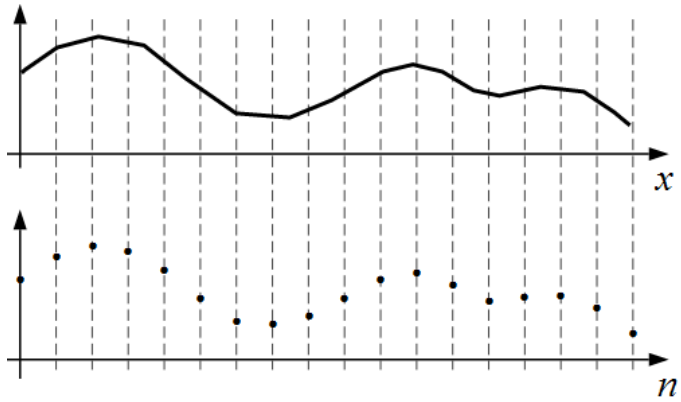
- Displayed primitives (lines, circles and curves) generated by the raster algorithms have a jagged, or stairstep, appearance because the sampling process digitizes coordinate points on an object to discrete integer pixel positions.
- This process causes distortion of information due to low-frequency sampling (undersampling) called aliasing
- Aliasing is said to be stair effect produced when drawing an object or a line.



Aliasing also forms a jagged right angle patterns at the edges of objects. The jaggies can be visually distracting, especially for high contrast edges near horizontal or vertical.

When we have a continuous signal to be represented, at times it becomes difficult to represent the whole infinite points of the signal.

Therefore the continuous signal can only be represented by sampling it at discrete points.



The sampled points are then used to *reconstruct* a new signal, that is intended to represent the original signal.

However, the reconstructed signals are a false representation of the original signals

In the English language, when a person uses a false name, that is known as an *alias*, and so it was adapted in *signal analysis* to apply to falsely represented signals

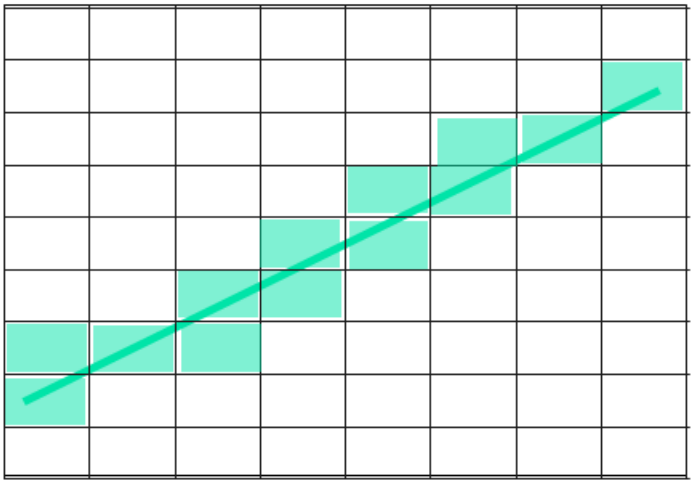
Aliasing in computer graphics usually results in visually distracting *artifacts*.

However in drawing a line or an object, the process of determining the pixel representation also produces a stairstep or jaggies like what we have in signal sampling.

All rasterized locations do not match the true line and thus we have to select the best candidate pixel for representation

Line Segments

- If we tried to sample a line segment so it would map to a 2D raster displa we would see stair steps, or jaggies since we “quantized” the pixel values to 0 or 1.



- The effect created when rasterization is performed over a discrete series of pixels.
- In particular, when lines or edges do not necessarily align directly with a row or column of pixels, that line may appear unsmooth and have a stair-step edge appearance.

Antialiasing

A lot of effort goes into trying to stop the stairstep appearence. This is known as *antialiasing.*

- Antialiasing utilizes blending techniques to blur the edges of the lines and provide the viewer with the illusion of a smoother line.
- Points, lines or polygons can be antialiased by increase sampling rate with raster systems by simply displaying objects at higher resolution.
- But even at the highest resolution possible with current technology, the jaggies will be apparent to some extent.

There are two general approaches:

- Area sampling and
- Supersampling.

Supersampling

A straightforward antialiasing method is to increase sampling rate by treating the screen as if it were covered with a finer grid than is actually available.

We can then use multiple sample points across this finer grid to determine an appropriate intensity level for each screen pixel.

This technique of sampling object characteristics at a high resolution and displaying the results at a lower resolution is called supersampling (or postfiltering, since the general method involves computing intensities ,it subpixel grid positions, then combining the results to obtain the pixel intensities).

Area sampling

An alternative to supersampling is to determine pixel intensity by calculating the areas of overlap of each pixel with the objects to be displayed. Antialiasing): by computing overlap areas is referred to as area sampling (or prefiltering, since the intensity of the pixel as a whole is determined without calculating subpixel intensities).

Pixel overlap areas are obtained by determining where object boundaries intersect individual pixel boundaries.

Area sampling approaches sample primitives with a box (or Gaussian, or whatever) rather than spikes

- Requires primitives that have area (lines with width)
- Sometimes referred to as pre-filtering

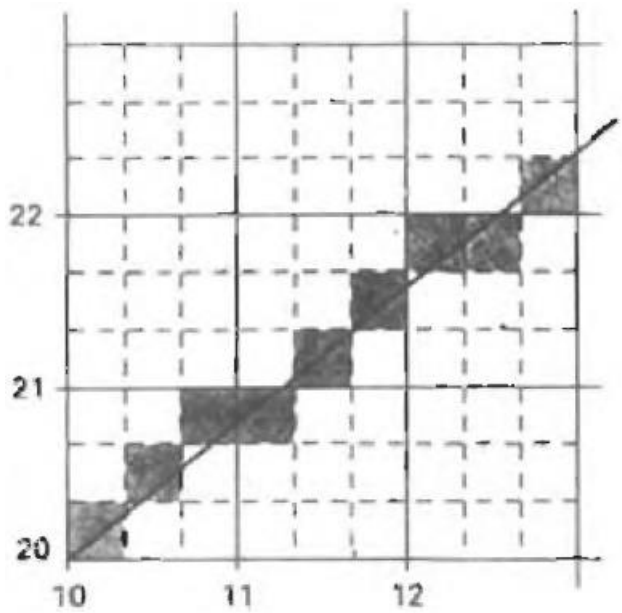
Supersampling Straight Line Segments

Supersampling straight lines can be performed in several wavs.

For the grayscale display of a straight-line segment, we can divide cach pixel into a number of subpixels and count the number of subpixels that are along the line path.

The intensity level for each pixel is then set to a value that is proportional to this subpxel count.

An example



Each square pixel area is divided into nine cqual-sized square subpixels, and the shaded regions show the subpixels that would be selected by Brescnhani's algorithm.

This scheme provides for three intensity settings above zero, since the maximum number of subpixels that can be selected within any pixcl is three.

For this example,

- the pixel at position (10. 20) is set to the maximum intensity (level 3);
- pixels at (11, 21) and (12,Zl) are each set to the next highest intensity (level 2); and
- pixels at (11, 20) and (12, 22) are each set to the lowest intensity above zero (level 1).

Thus the line intensity is spread out over a greater nuniber of pixels, and the stairstep effect is smoothed by displaying a somewhat blurred line path in the vicinity of the stair step. (between horizontal runs).

Pixel-Weighting Masks

Supersampling algorithms are often implemented by giving more weight to subpixels near the center of a pixel area, since we would expect these subpixels to be more important in determining the overall intensity of a pixel.

For the 3 bv 3 pixel subdivisions, a weighting scheme is as follows

1	2	1
2	4	2
1	2	1

The center subpixel is weighted four times that of the corner subpixels and twice that of the remaining subpixels. Intensities calculated for each grid of nine subpixels would then be averaged so that:

- the center subpixel is weighted by a factor or 1/4;

- the top, bottom, and side subpixels are each weighted by a factor of 1/8; and
- the corner subpixels are each weighted by a factor of 1 /16.
-

An atray of values specifying the relative importance of subpixels is sometimes referred to as a "mask" of subpixel weights. Similar masks can be set up for larger subpixel grids.

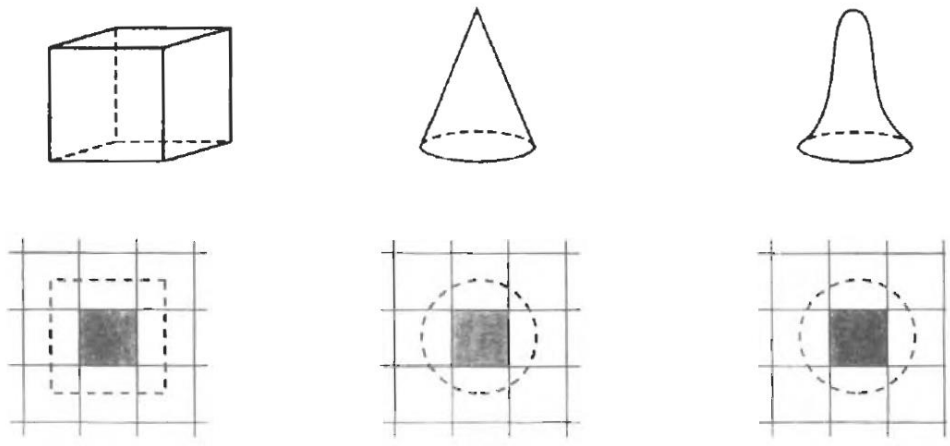
Also, these masks are often extended to include contributions from subpixels belonging to neighboring pixels, so that intensities can be averaged over adjacent pixels.

Pixel phasing

- Shifting the display location of pixel areas
- Applied by micropositioning the electron beam in relation to object geometry.

Filtering techniques

- More accurate method for antialising lines
- Method similar to applying weighted pixel mask, but now we imagine a continuous weighing surface covering the pixel.
- Rectangular, cone or gaussian filters (mask)



Computer Animation

Animation means giving life to any object in computer graphics.
It has the power of injecting energy and emotions into the most seemingly inanimate objects.

Computer-assisted animation and computer-generated animation are two categories of computer animation. It can be presented via film or video.

The basic idea behind animation is to play back the recorded images at the rates fast enough to fool the human eye into interpreting them as continuous motion.

Animation can make a series of dead images come alive.
Animation can be used in many areas like entertainment, computer aided-design, scientific visualization, training, education, e-commerce, and computer art.

Animation Techniques

Traditional Animation (frame by frame)

Traditionally most of the animation was done by hand. All the frames in an animation had to be drawn by hand. Since each second of animation requires 24 frames (film), the amount of efforts required to create even the shortest of movies can be tremendous

Key-framing

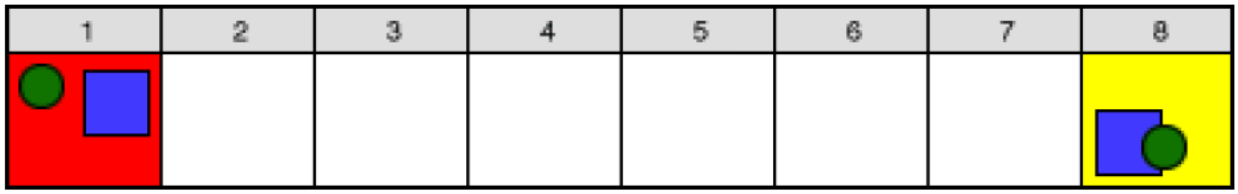
In this technique, a storyboard is laid out and then the artists draw the major frames of the animation. Major frames are the ones in which prominent changes take place.

They are the key points of animation. Key-framing requires that the animator specifies critical or key positions for the objects.

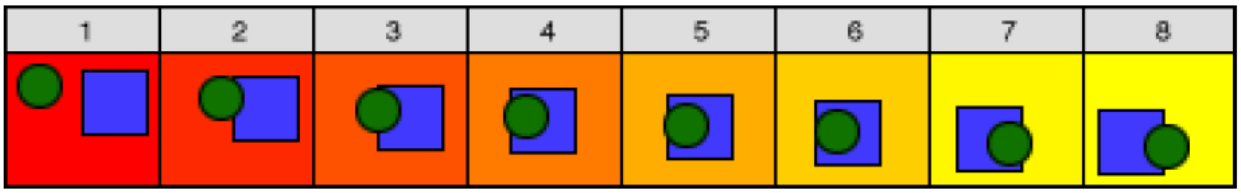
The computer then automatically fills in the missing frames by smoothly interpolating between those positions.

A key-frame is a frame where we define changes in animation. Every frame is a key-frame when we create frame by frame animation. When someone creates a 3D animation on a computer, they usually don't specify the exact position of any given object on every single frame.

They create key-frames. Key-frames are important frames during which an object changes its size, direction, shape or other properties. The computer then figures out all the in-between frames and saves an extreme amount of time for the animator. The following illustrations depict the frames drawn by user and the frames generated by computer.



Frames drawn by user



In between frames drawn by the computer.

Procedural
In a procedural animation, the objects are animated by a procedure - a set of rules - not by key-framing. The animator specifies rules and initial conditions and runs simulation. Rules are often based on physical rules of the real world expressed by mathematical equations.

Behavioral
In behavioral animation, an autonomous character determines its own actions, at least to a certain extent. This gives the character some ability to improvise, and frees the animator from the need to specify each detail of every character's motion.

Performance Based (Motion Capture)
Another technique is Motion Capture, in which magnetic or vision-based sensors record the actions of a human or animal object in three dimensions. A computer then uses these data to animate the object.

This technology has enabled a number of famous athletes to supply the actions for characters in sports video games. Motion capture is pretty popular with the animators mainly because some of the commonplace human actions can be captured with relative ease. However, there can be serious discrepancies between the shapes or dimensions of the subject and the graphical character and this may lead to problems of exact execution.

Physically Based (Dynamics)
Unlike key framing and motion picture, simulation uses the laws of physics to generate motion of pictures and other objects. Simulations can

be easily used to produce slightly different sequences while maintaining physical realism.

Secondly, real-time simulations allow a higher degree of interactivity where the real person can maneuver the actions of the simulated character.

In contrast the applications based on key-framing and motion select and modify motions form a pre-computed library of motions. One drawback that simulation suffers from is the expertise and time required to handcraft the appropriate controls